

Table S1. SQUID simulation parameters

parameter	symbol	value	unit
Thermal power	P_th	3400.0	MW
Core gamma flux (RPV inner)	Phi_gamma	4500000000000.0	cm ⁻² s ⁻¹
Fast neutron flux (RPV)	Phi_n	120000000000.0	cm ⁻² s ⁻¹
SQUID loop area	A_sq	10000.0	um ²
Loop inductance	L_sq	80.0	pH
Critical current (fresh)	Ic0	10.0	uA
Junction capacitance	C	2.0	fF
Normal resistance	R_n	5.0	Ohm
Operating temperature	T	4.2	K
Array size	N	64.0	-
Dephasing time (fresh)	T2_0	8.0	us
VQC qubits	n	8.0	-
VQC layers	L	4.0	-

Table S2. Material / radiation-damage parameters (Nb)

	parameter	value	unit
	Damage coeff alpha_I (Nb)	1.2e-20	cm ²
	Damage coeff kappa_TLS (Nb)	5e+20	cm ⁻²
	1/f noise coeff fresh A_phi0 (Nb)	2.5e-05	Phi0 ² /Hz
Rad-induced dephasing coeff alpha_phi (Nb)		1e-09	cm ² s ⁻¹
Fresh dephasing rate Gamma_deph0/2pi (Nb)		20.0	kHz
NRT displacement energy Ed (Nb)		60.0	eV
arc-dpa recombination factor xi (Nb)		0.3	-
Displacement cross-section sigma_dpa (Nb)		600.0	mb

Table S3. Noise-spectrum summary by lifecycle stage

Lifecycle stage	N snapshots x freq bins	mean	std	min	max
2yr	5120	1.786e+04	2.819e+04	105.6	1.362e+05
5yr_precursor	5120	2.232e+04	3.521e+04	132.7	1.71e+05
7yr_prefail	5120	2.428e+04	3.839e+04	147	1.935e+05
fresh	5120	7308	1.154e+04	42.59	5.742e+04

Table S4. Sensor lifecycle-trace milestones

Years	DDD (dpa)	Ic (uA)	A_phi	f_c (Hz)	QFI (N=64)	Visibility
0	0	10	2.5e-05	9.0014e+07	4031	0.99203
1.005	6.8507e-07	9.864	0.00010674	3.8432e+08	3499.9	0.92437
2.0101	1.3701e-06	9.7406	0.00014887	5.36e+08	3038.8	0.86133
3.0151	2.0552e-06	9.6273	0.00018146	6.5337e+08	2638.4	0.80259
4.0201	2.7403e-06	9.5221	0.00020904	7.5266e+08	2290.8	0.74785
5.0251	3.4253e-06	9.4236	0.00023338	8.4029e+08	1989	0.69684
6.0302	4.1104e-06	9.3306	0.00025541	9.1961e+08	1726.9	0.64932
7.0352	4.7955e-06	9.2425	0.00027568	9.9262e+08	1499.4	0.60503
8.0402	5.4806e-06	9.1584	0.00029457	1.0606e+09	1301.9	0.56377
9.0452	6.1656e-06	9.078	0.00031231	1.1245e+09	1130.3	0.52532
10	6.8164e-06	9.0044	0.00032828	1.182e+09	988.36	0.49122

Table S5. QFI vs. array size N (fresh sensor, t=0)

N	QFI	visibility
1	0.99975	0.999875
2	3.998	0.99975
4	15.984	0.9995
8	63.8721	0.999
16	254.978	0.998002
32	1015.84	0.996008
64	4030.99	0.992032
128	15868	0.984127
256	61473.1	0.968507

Table S6. Anomaly-detection performance by failure mode

failure_mode	TPR_quantum	TPR_classical_p	AUC_quantum	AUC_classical_p	FPR_quantum	FPR_classical_x1
Sensor degradation	100	100	1	1	10	10
Radiation burst	100	100	1	1	10	10
Coolant flow anomaly	15.2	33.8	0.7457	0.7539	10	10
Foreign material	100	100	1	1	10	10
Calibration drift	96.8	98.4	0.9972	0.9961	10	10

Table S7. Early-warning horizon by failure mode

	failure_EW_detected	quantum_EW_detected	quantum_hours	detected_EW_and	classical_EW_detected	classical_hours	detected_classical
Sensor degradation		48.48	3.89	25	56.88	4.274	25
Radiation burst		12.32	4.514	25	15.44	4.784	25
Coolant flow anomaly		7.455	4.6	11	21.48	6.439	23
Foreign material		28.56	4.614	25	28.8	4.561	25
Calibration drift		18.08	7.008	25	20.08	7.364	25

Table S8. VQC hyperparameter sweep (full results)

L	n	lr	AUC_mean	AUC_std	AUC_95pct_thr	AUC_95pct_thr	n_seeds
2	8	0.01	0.95431	0.0011742	0.053333	0	3
3	8	0.01	0.94627	0.00010887	0.053333	0	3
4	8	0.01	0.9476	0.0035652	0.053333	0	3
5	8	0.01	0.94618	3.6289e-05	0.053333	0	3
4	6	0.01	0.95052	0.0081047	0.053333	0	3
4	10	0.01	0.94545	0.0036173	0.053333	0	3
4	8	0.05	0.95376	0.0031201	0.053333	0	3
4	8	0.001	0.94532	0.0014334	0.053333	0	3

Table S9. Numerical-accuracy validation-suite results

	Check	Result	Detail
Independent, first-principles recomputation checks (module_02_validation_suite.py).	QFI closed-form recomputation (N=64, 5 spot channels)	PASS	max relative error = 1.15e-16
	Entanglement-visibility identity $V=\sqrt{\text{QFI}}/N$ (all N=64)	PASS	max abs error = 2.22e-16
	ROC curve TPR monotonicity (non-decreasing with FPR, 1000 points)	PASS	no violations
	Recomputed overall AUC lies in valid [0,1] range (both quantum and classical)	PASS	quantum AUC=0.9486; classical AUC=0.9500
	er-frequency identity $f_c=A_{\phi}/\text{floor}$ implies constant floor (2000 points)	PASS	relative discrepancy = 6.45e-13
	All probability/fraction/rate columns within documented valid range	PASS	
	Anomaly-scores table has 5 distinct failure modes + healthy baseline	PASS	healthy n=20000, anomaly n=2500, modes=5
	Failure-mode labels by matches between normality_scores and anomaly_scores	PASS	radiation_burst, 'sensor_degradation', anomaly_scores=['calibration_drift', 'scintillator_normality', 'foreign_material', 'radiation_burst', 'sensor_degradation'], anomaly_scores=['calibration_drift', 'scintillator_normality', 'foreign_material', 'radiation_burst', 'sensor_degradation']